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Effects of Surface Micro-Geometry on the Pressures and Internal Stresses of Pure Rolling EHL Contacts[©]

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A fast approach for the calculation of 3-D pressures and sub-surface stresses in pure rolling EHL contacts with real (measured) roughness is presented. For the calculation of pressures the model uses the analytical work of Greenwood and Morales-Espejel for transverse (to the rolling direction) moving sinusoidal waviness. The model is extended to handle longitudinal waviness as well. By using these two basic components the pressure distribution for real rough surfaces can now be calculated from the 3-D roughness map. From the surface pressures, the sub-surface stress field can then be calculated with a similar technique. The result is the total stress state in a rolling rough EHL contact. In this paper the method for calculating the pressures and stress fields is first described. It is then applied to several types of surface finish.

KEY WORDS

Elasto-Hydrodynamic Lubrication; Surface Roughness Effects

INTRODUCTION

In recent years the understanding of the behaviour of roughness in elastohydrodynamic lubrication (EHL) has gained great importance in industry. The demands of higher performance and mechanical components operating with thinner lubricating films have supported research in this area.

Real roughness can be seen as the superposition of waviness with crests running in both directions, transverse (to the contact rolling direction) and longitudinal (along the contact rolling direction). Time dependent effects occur in the contact if there are

transverse components and this roughness orientation has recently received much attention.

Many publications have contributed to the understanding of EHL with rough surfaces particularly with transverse orientation. The first analysis of a surface feature moving into an EHL contact was due to Lee and Cheng (1) and concluded that the elastic deformation of the features is significant. An important step was later achieved with perhaps the first numerical solution of the EHL problem with moving roughness, due to Chang, et al. (2). Surprisingly the analysis included non-Newtonian fluids. This early work already revealed the complex kinematic behaviour of surface deformation and pressure generation present in EHL with moving roughness. Their main conclusion was that the squeeze term in the Reynolds equation significantly changes the local pressure generation and therefore surface deformation. Other important work was due to Venner (3) who also solved numerically the EHL problem of a rough surface moving over a smooth one and exhibited for the first time the apparent disassociation of pressures and the surface features that produced them, specially when large sliding is present. He observed that the pressure fluctuations travel with the velocity of the rough surface while the film thickness variations travel with the mean velocity of the lubricant. Venner also speculated that in the high pressure zone of the contact, the lubricant viscosity can be so large that the influence of the Poiseuille term in the Reynolds equation simply vanishes, reducing it to the transport equation, which is linear.

Morales-Espejel (4) and Greenwood and Morales-Espejel (5) showed that Venner's observations can be explained by considering just the inlet of the contact. In this region, the pressure is low and so is the roughness deformation; therefore each roughness peak entering the contact will behave as a flow exciter, closing and opening the inlet to allow less or more fluid to enter. This inlet excitation (complementary function) has to be added to the stationary EHL solution (particular integral) to produce the complete EHL transient solution. However, Greenwood and Morales-

NOMENCLATURE

	= half-width of the elliptical Hertzian contact, rolling direction	[m]	$\mathbf{v}$	= elastic displacements	[m]
A	= constant $A = 2\lambda/(\pi E')$	[m/Pa]	x	= rolling direction coordinate, measured from edge of contact	[m]
A_d	= dimensionless deformed waviness amplitude	[-]	X	= dimensionless rolling direction coordinate, $X = x/a$	[-]
A_i	= dimensionless initial waviness amplitude	[-]	y	= transverse direction coordinate	[m]
b	= half-width of the elliptical Hertzian contact, transverse direction	[m]	Y	= dimensionless transverse direction coordinate, $Y = y/a$	[-]
C	= compressibility constant $C = (\gamma_1 - \beta_1) \bar{h}$	[m/Pa]	z	= vertical direction coordinate, roughness height	[m]
C_0	= dimensionless compressibility term $C_0 = \frac{\pi E'}{2a} M^{0.5} L^{-0.5} (\gamma_1 - \beta_1) \bar{h}$	[-]	Z_l	= dimensionless waviness amplitude, $Z_l = z_l R_x / a^2$	[-]
E'	= elasticity modulus of body i $E' = \frac{1}{2} \left(\frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2} \right)^{-1}$	[Pa]	α	= pressure-viscosity coefficient	[Pa ⁻¹]
E_c	= dimensionless constant $E_c = \frac{E' a}{2 p_H R_x}$	[-]	β	= constant in the density equation, $\beta = 1.683 \times 10^{-9}$	[Pa ⁻¹]
ϵ	= full elliptical integral of the 2nd kind	[-]	β_l	= constant in the compressibility term, $\beta_l = \beta/(1 + \beta p_H)$	[Pa ⁻¹]
f	= frequency, $f = \omega/2\pi$	[m ⁻¹]	γ	= constant in the density equation, $\gamma = 2.266 \times 10^{-9}$	[Pa ⁻¹]
F	= contact load	[N]	γ_l	= constant in the compressibility term, $\gamma_l = \gamma/(1 + \gamma p_H)$	[Pa ⁻¹]
h	= film thickness	[m]		= correction factor $\epsilon = \frac{\pi^2 k (1-k^2)^{3/2}}{16(\epsilon - k^2 K^2)}$ in elliptical contacts	[-]
$\bar{h}$	= mean film thickness	[m]	η	= lubricant viscosity	[Pa s]
H	= dimensionless film thickness, $H = h R_x / a^2$	[-]	η_0	= lubricant viscosity at ambient pressure	[Pa s]
$H_{l,2}$	= dimensionless film thickness amplitude, particular integral, complementary function	[-]	θ	= phase shift of the convected wave	[rad]
k	= ellipticity ratio $k = \text{olh}$	[-]	ϕ	= waviness orientation angle, relative to y-axis	[rad]
K	= full elliptical integral of the 1st kind	[-]	λ	= waviness wavelength	[m]
L	= material parameter (circular contacts), $L = \alpha E' \left(\frac{\eta_0 2a}{E' R_x} \right)^{1/4}$	[-]	ν_i	= Poisson ratio of body i	[-]
M_c	= load parameter (circular contacts), $M_c = \frac{F}{E' R_x^2} \left(\frac{E' R_x}{2 \eta_0 a} \right)^{3/4}$	[-]	ρ	= lubricant density	[kg m ⁻³]
M_e	= load parameter (elliptical contacts), $M_e = \epsilon M_c$	[-]	ρ_0	= lubricant density at atmospheric pressure	[kg m ⁻³]
p	= pressure	[Pa]	σ	= a normal stress	[Pa]
p_H	= maximum Hertzian pressure	[Pa]	σ_{vM}	= von Mises stress in tension or compression	[Pa]
P	= dimensionless pressure, $P = p/p_H$	[-]	τ	= shear stress	[Pa]
$P_{l,2}$	= dimensionless pressure amplitude, particular integral, complementary function	[-]	τ_{vM}	= von Mises shear stress	[Pa]
	= reduced radius of curvature. $R' = \left(\frac{1}{R_x} + \frac{1}{R_y} \right)^{-1}$	[m]	ω	= angular frequency, $\omega = 2\pi/\lambda$	[rad m ⁻¹]
t	= time	[s]	∇	= dimensionless wavelength parameter, $V = (\lambda/a) M_e^{0.5} L^{-0.5}$	[-]
T	= dimensionless time, $T = \bar{u}t/a$	[-]	$l,2$	= lower, upper surface	[-]
u_1	= velocity, lower surface (smooth)	[m s ⁻¹]	x,y,z	= in x, y- or z-direction	[-]
u_2	= velocity, upper surface (rough)	[m s ⁻¹]	$\bar{g}$	= mean of g	[-]
$\bar{u}$	= mean velocity, $\bar{u} = (u_1 + u_2)/2$	[m s ⁻¹]			

Espejel were not able to obtain a solution for the amplitude of the excitation function. Their model was based only on the solution of the Couette and squeeze terms (linearised Reynolds equation) which is valid only in the high pressure zone.

Using full numerical solutions Venner, et al. (6) showed that for transverse sinusoidal waviness the ratio between the amplitudes of the original and deformed film thickness fluctuations is determined by a single dimensionless variable V , producing a universal amplitude reduction curve. Later Venner and Lubrecht (7) extended this concept to any orientation of the harmonic pattern in pure rolling and for elliptical contacts.

Greenwood (8) and Morales-Espejel, et al. (9) used the new amplitude reduction curve to estimate the missing amplitude of the excitation function and with the combination of Fourier analy-

sis they were able to predict film thickness and pressure distributions of contacts with **any** moving transverse roughness or features in **any** slide-roll combination, with Newtonian fluid.

However, the complete analytical solution of the problem is due to Hooke (10). He extended a limiting solution for smooth EHL contacts well in the elastic piezoviscous regime to cases with waviness in the inlet and he was able to predict both complete pressure and film thickness distributions for moving transverse roughness. Hooke (11) further extended his model to account for compressibility and even non-Newtonian behaviour, Hooke (12). However, the extension requires numerical schemes for its solution. Despite this important step due to Hooke, a compressible closed-form solution is needed to speed-up the solution time in 3-D real roughness problems.

One important practical application of the Venner, et al. amplitude reduction curve is presented by Masen, et al. (13) where the curve is used in combination with Fourier decomposition to predict the 3-D film thickness map i.e. the deformed roughness in the central zone of pure rolling elliptical contacts and the associated surface "lift-off curve." This technique enables engineers to rank surfaces by their ability to build-up film thickness. However, when the surface is deformed, pressure and therefore subsurface stress fluctuations are generated, which should also be considered in the ranking of surfaces. Presently, the only way to do this is by performing full EHL numerical solutions and multi-integration calculations of stresses which, despite the progress in algorithms and computer speed, is still impractical in real applications. One feasible alternative is the combination of simple analytical solutions with Fourier techniques.

An earlier attempt to calculate pressures fluctuations due to roughness also using Fourier techniques and a fast scheme is due to Hu, et al. (14). They used about 50 numerical simulations with wavy surfaces to obtain a curve-fit scheme that they applied to real surfaces by using Fourier decomposition. For 3-D isotropic roughness, however, the error becomes substantial when the results are compared with results from numerical solutions. The approach is time independent, valid only for pure sliding cases with stationary roughness.

In the calculation of sub-surface stresses an important step was given in Lubrecht and Ioannides (15) with the application of multi-level multi-integration.

The contribution of the present work is the development of a general approach to calculate EHL pressure fluctuations resulting from 3-D surfaces in pure rolling, based on the linearisation of the Reynolds equation in the central zone due to the high viscosity. The existing analytical solution for transverse sinusoidal waviness is used and a solution for longitudinal sinusoidal waviness is incorporated into the scheme. Next, by using Fourier analysis, full 3-D pressure distributions for the central zone of rolling EHL elliptical contacts are calculated. The sinusoidal components of the pressures are then used to calculate analytically the 2-D and 3-D subsurface stress field, following the scheme by Tripp, et al. (22). The work of Greenwood and Morales-Espejel (5) and Masen, et al. (13) is used for the film thickness distributions.

The model represents a simple, fast and efficient tool to rank surfaces for their ability to build up lubricant film and to constrain the interior stress field. The model has been validated using full numerical solutions for sinusoidal and irregular pattern surfaces. The agreement vs. full numerical solutions is good.

MATHEMATICAL MODEL FOR PRESSURES

Due to difficulties in geometry, analytical solutions for wavy surfaces in EHL presently exist only for line contacts. However, real life applications frequently involve circular and elliptical contacts. Since the present scheme intends to predict pressures well inside the contact (in the high pressure zone) far away from side-flow and low pressure areas, it is believed that an extension of line contact solutions is possible. This will be described in the following sections.

EHL Theory for Line Contacts

Solving the EHL problem with 3-D moving roughness using a numerical approach is a complex task. The main challenge is the large number of points needed to represent the roughness accurately. Since the problem is time dependent, the computer time needed for a single solution is impractical, specially for everyday engineering applications. Therefore, a different approach to the problem is needed.

In this paper the solution will be split by using the linearised approach for infinitely long line contacts having waviness with transverse or longitudinal orientation. Fourier decomposition techniques will be used for the surface. Subsequently for every sinusoidal component of roughness the corresponding pressure wave is found. Finally, pressures and deformed waves are re-assembled.

Starting with a line EHL contact, the problem is governed by the transient Reynolds equation for a compressible Newtonian fluid:

$$\frac{\partial}{\partial x} \left(\frac{\rho h^3}{12\eta} \frac{\partial p}{\partial x} \right) = \bar{u} \frac{\partial \rho h}{\partial x} + \frac{\partial \rho h}{\partial t} \quad [1]$$

where the film thickness $h = h(x, t)$,

$$h(x, t) = h_o + z(x, t) + \frac{x^2}{2R}$$

$$- \frac{4}{\pi E'} \int_{-\infty}^{\infty} p(s) \ln|x - s| ds \quad [2]$$

is given as the sum of the integration constant, h_o , the initial roughness, $z(x, t)$, the curvature of the contact, $x^2/2R$ and the elastic displacement due to the pressures.

The load balance equation is given by:

$$F = \int_{-\infty}^{\infty} p(x) dx \quad [3]$$

the Barus law is used to describe the viscosity-pressure relationship:

$$\eta(p) = \eta_0 e^{\alpha p} \quad [4]$$

and the Dowson and Higginson relationship is used to relate the lubricant density to the pressure:

$$\frac{\rho(p)}{\rho_o} = \frac{1 + \gamma p}{1 + \beta p} \quad [5]$$

where $\beta = 1.683 \times 10^{-9} \text{Pa}^{-1}$, $\gamma = 2.266 \times 10^{-9} \text{Pa}^{-1}$ and ρ_o is the lubricant density at atmospheric pressure.

Reduction of the Reynolds Equation for the Central Zone of the Contact

Greenwood (8) explored the conditions in which the Reynolds equation can be reduced to the simple transport equation; summarised here. In order to justify the use of the Hertzian pressure distribution, Grubin in his solution for a smooth surface contact

stated that the pressures in the lubricated contact must be of the same order as the pressures in the same dry contact, typically 1 GN/m^2 . The pressure gradients are large, $(1 \times 10^9 \text{ N/m}^2)/(1 \times 10^{-4} \text{ m})$, but the film thickness is small ($1 \times 10^{-6} \text{ m}$) which largely cancels it. Therefore, the ratio of the pressure flow $[\rho h^3/(12\eta)]$ to the shear flow $\bar{u}(\rho h - \rho^* h^*)$ is of order $1/\eta$. Since the viscosity is very large, which means $\bar{u}(\rho h - \rho^* h^*) \approx 0$, for an incompressible fluid Grubin correctly concluded that the central zone of the lubricated contact was largely parallel.

In a rough surface contact, the argument can be applied for a sinusoidal pressure variation $p = p_1 \cos(2\pi x/\lambda)$ the greatest pressure gradient is $(2\pi p_1/\lambda)$. Therefore, the steady-state form of Eq. [1] with an incompressible fluid,

$$\left[\frac{h^2}{12\eta\bar{u}} \right] \left(\frac{dp}{dx} \right) = \frac{h - h^*}{h}$$

will have either term of the order $(\pi h^2/6\eta\bar{u})(p_1/\lambda)$. If the h variation in the central zone is to be less than 1% and the mean film thickness is no larger than $(h \approx 1 \times 10^{-6} \text{ m})$ and with typical $0.04 \geq \eta_0 \bar{u} \leq 0.1 \text{ N/m}$, the condition for the linearisation is about,

$$(\eta_0 \eta)(p_1/\lambda) < 5 \times 10^8$$

For $\alpha = 20 \times 10^{-9} \text{ m}^2/\text{N}$ and pressure variations of amplitude 0.5 GPa and $\lambda = 50 \times 10^{-6} \text{ m}$, the minimum pressure should not exceed 0.5 GPa .

In a real surface, many of the waves are in this situation but combinations of short wavelength and high amplitudes are not. Especial care must be taken in those unfortunate situations.

Transverse Waviness: Simplifications for the High Pressure Zone

For transverse waviness, it is important to describe the main aspects given by Greenwood and Morales-Espejel (5) since this is the basis of the present work. For the moment the condition of pure rolling, $u_1 = u_2$ will be relaxed.

For typical operating conditions it has been established that the left hand side of Eq. [1] becomes very small compared to the terms on the right hand side. This allows linearisation of the Reynolds equation to the following transport equation,

$$\bar{u} \frac{dph}{dx} + \frac{\partial \rho h}{\partial t} = 0 \quad [6]$$

which can be rewritten as,

$$\bar{u} \frac{\partial r}{\partial x} + \frac{\partial r}{\partial t} = 0 \quad [7]$$

where the oil mass flow per unit of width, ph , is replaced by

$$r = \frac{\rho h}{\rho^* h^*} \quad [8]$$

this linearisation is the crucial step in the present approach.

The upper surface is moving with a velocity u_2 and with sinusoidal waviness,

$$z = z_1 \cos[\omega(x - u_2 t)] \quad [9]$$

where $\omega = 2\pi/\lambda$ (frequency). The lower surface (smooth) is moving with velocity u_1 .

Solutions for r , satisfying Eq. [7], can easily be found. The complete solution is made of a particular integral, $r_i = 1$, and a complementary function, $r_c = f(x - Gt)$.

Particular Integral: Steady-State

The particular integral for Eq. [8] represents the solution of the equation in pure sliding (stationary roughness and moving smooth surface): $r_i = 1$. So $ph = p^* h^*$, where p^* and h^* should be values of the fluid density and film thickness occurring simultaneously. Hence, p^* may be taken as the density at the mean pressure $\rho(p_H) = \rho_H$ and h^* as the film thickness occurring at this pressure. Greenwood and Morales-Espejel (5) have shown that by linearising the Dowson and Higginson Eq. [5], the density can be related to the pressure as follows

$$\frac{\rho(p)}{\rho(p_H)} \approx 1 + (\gamma_1 - \beta_1)(p - p_H) \quad [10]$$

where $\beta_1 = \beta/(1 + \beta p_H)$ and $\gamma_1 = -\gamma/(1 + \gamma p_H)$. The linearisation holds for "small" values of $p - p_H$, thus small roughness amplitudes.

They found that the amplitude of the pressure fluctuations for this component is given by,

$$p_1 = \frac{-z_1}{C + A} \quad [11]$$

where

$$A = \frac{2\lambda}{\pi E'}, \quad C = (\gamma_1 - \beta_1) \bar{h}$$

The pressure fluctuations are moving with the velocity u_2 , so

$$\Delta p_i(x) = p_i(x) - p_H = p_1 \cos[\omega(x - u_2 t)] \quad [12]$$

Notice that the pressure amplitude is determined by the relative compressibility of the lubricant and the flexibility of the surface material.

Similarly, the film thickness variation is given as,

$$h_i(x) = h_1 \cos[\omega(x - u_2 t)] \quad [13]$$

with

$$h_1 = \frac{z_1 C}{C + A} \quad [14]$$

Complementary Function: Inlet Excitation

The solution of the complementary function is far more complex, again Greenwood and Morales-Espejel (5) is followed.

At the inlet, the pressures are low and do not deform the roughness completely, so when the partially deformed amplitudes enter into the high pressure zone they will produce a fluctuation (in time) of lubricant flow that will be transported down stream with the velocity of the lubricant $\bar{u}$, say $r_c = f(x - \bar{u}t)$. Since the initial roughness is sinusoidal, one can also expect a sinusoidal function

for the flow fluctuation. Thus $r_c = r_2 \cos[\omega'(x - \bar{u}t) + \theta]$ satisfies Eq. [7].

θ now appears as the phase-shift between the film thickness variation and the original roughness, as introduced by Hooke.

Therefore

$$h_c = h_2 \cos[\omega'(x - \bar{u}t) + \theta] \quad [15]$$

$$\Delta p_c = p_2 \cos[\omega'(x - \bar{u}t) + \theta] \quad [16]$$

For pure rolling the phase shift is almost zero, therefore it will be disregarded in the pure rolling calculations, although retained in the general case.

Greenwood and Morales-Espejel (5) also show that the pressure amplitude is related directly to the film thickness amplitude by,

$$p_2 = \frac{h_2}{C + A'} \quad [17]$$

where θ as yet will remain unknown, and $\omega' = 2\pi/\lambda'$ and $\lambda' = \lambda \bar{u}/u_2$, therefore $A' = 2\lambda'/(\pi E')$. This shows that the wavelength of the convected wave is a function of the sliding velocity. Defining the slide/roll ratio as $S = (u_1 - u_2)/\bar{u}$, then $\bar{u}/u_2 = 1/(1 - S/2)$ and therefore $\omega' = \omega(1 - S/2)$. For pure rolling, however, $\bar{u} = u_2$, therefore $\lambda' = \lambda$ and $A' = A$.

The complete solution will be obtained by adding the two components $r = r_i + r_c$ and $\Delta p = \Delta p_i + \Delta p_c$, with the inlet excitation amplitude Δh still unknown. The total pressure fluctuation amplitude due to transverse waviness is,

$$\Delta p = \frac{h_2}{C + A'} \cos[\omega(1 - S/2)(x - \bar{u}t) + \theta] - \frac{z_1}{C + A'} \cos[\omega(x - u_2 t)] \quad [18]$$

Notice that Eq. [18] shows the two pressure components. In solutions where sliding is present, frequently the particular integral dominates over the complementary function. However, for the film thickness it is in general the opposite. This explains the apparent disassociation of pressure and film thickness shown in many numerical solutions.

Hooke (10) applied a similar analysis, although incompressible. By extending a smooth surface limiting solution scheme, Hooke is also able to obtain a solution for the h_2 term and the phase-shift θ .

Combining Eqs. [33] and [60] from Hooke (10) it is possible to obtain the final pressures inside the contact. This expression can further be reduced by assuming large x (away from the inlet). Then the terms g_2 and f_2 given in the reference will die out quickly leading to,

$$\dot{P}(\bar{x}) = \frac{2\bar{\omega}}{3} \bar{a} \cos[(1 - S/2)\bar{\omega}\bar{x} + \phi + \theta] - \frac{2\bar{\omega}}{3} \bar{A} \cos[\bar{\omega}\bar{x} + \phi] \quad [19]$$

where $\phi = -(1 - S/2)\bar{\omega}\bar{t}$.

Hooke used the following dimensionless groups, $\dot{P} = \frac{P}{E'} G^{0.25}$, $\bar{A} = \frac{3}{8} k z_0 G^{0.75}$, $\bar{a} = \frac{3}{8} k (\Delta h) G^{0.75}$,

$\bar{\omega} = \frac{\omega}{k G^{0.25}}$, $\bar{x} = k x G^{0.25}$, $\bar{t} = k \bar{u} t G^{0.25}$ and the dimensionless constants,

$$k = \frac{1}{\sqrt{8\pi}} \frac{1}{R'} \left[\frac{W}{E' R'} \right], \quad G = \frac{16}{27} \frac{1}{\alpha \eta_0 \bar{u} k}$$

Equation [19] (from Hooke) can be written as,

$$\Delta p = \frac{h_2}{A'} \cos[\omega(1 - S/2)(x - \bar{u}t) + \theta] - \frac{z_1}{A} \cos[\omega(x - u_2 t)] \quad [20]$$

Equation [20] is basically the same as Eq. [18], but without the compressibility term (C). Equations for h_2 and θ are given in Hooke (10). By using this scheme the complete solution for pressure and film thickness can be obtained analytically.

Finally, a note about the linearisation used for the density equation. In order to be accurate, such linearisation requires the pressure amplitudes to be small, although the definition of "small" is not precise, because it depends on p_H . However, with $p_H = 2$ GPa the linearisation works well for $0.5 \leq \Delta p \leq 4$. If needed once the pressures have been obtained, a correction can be introduced to the compression constant C_0 to re-calculate corrected pressures. The constant C, will need to be multiplied by the factor:

$$k_c = [\rho(p)/\rho(p_H) - 1] \frac{1}{C_0(p - p_H)}$$

$$k_i = [1 - \rho(p_H)/\rho(p)] \frac{1}{C_0(p - p_H)}$$

with k_i being the correction constant in the particular integral component and k_c in the complementary function component. The factors are different, because in the first case the linearised quantity is $\rho(p_H)/\rho(p)$, and in the second case it is the reciprocal.

The Extension to Elliptical Contacts

Hooke and Venner (16) show that there is a single underlying mechanism controlling the behaviour of roughness in all EHL contacts. For a set of comparable conditions they obtained numerical results for amplitude attenuation of waviness in point and line contacts. Slight differences that do exist between the two types of contact are due to side flow effects in point contacts and differences in elastic behaviour of the two conjunctions. Hooke and Venner thus have proved that the use of the line contact scheme for transverse roughness can immediately be extended to point contacts if the observations are done in the center of the contact (away from side-flow, elasticity and low pressure effects).

Pressure and film thickness amplitudes along the center-line of a circular contact have already been successfully predicted using line-contact theory, Greenwood (8).

Now the extension from circular to elliptical contacts is also straightforward as explained by Venner, et al. (17), the only requirement to be added is that the comparisons should be made using the same maximum Hertzian pressure. This needs the replacement of the circular contact parameter M_c with the more general elliptical contact parameter $M_e = \epsilon M_c$. Where ϵ is a correction coefficient described in the nomenclature.

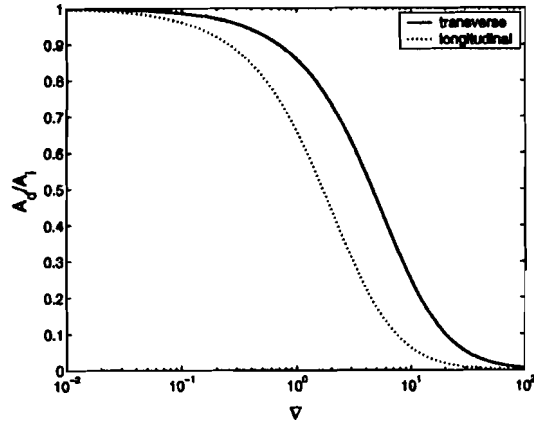


Fig. 1—Venner and Lubrecht's amplitude reduction curves for circular contacts and pure rolling.

Therefore, Eq. [18] can be written in dimensionless form and scaled using dimensionless parameters for either circular or elliptical contacts. The following dimensionless groups will be used; $X = x/a$, $T = \bar{u}t/a$, $Z = zR_x/a^2$, $H = hR_x/a^2$, $P = p/p_H$. For the wavelength the variable $V = (\lambda/a)M^{0.5}L^{-0.5}$ for point contacts, introduced by Venner and Lubrecht (7), will be used. Thus, Eq. [18] in pure rolling can be rewritten as,

$$M^{-0.5}L^{0.5}E_c^{-1}\frac{\Delta P}{Z_1} = \frac{\pi}{C_0 + \nabla} \left(\frac{H_2}{Z_1} \cos\left[\frac{2\pi}{\lambda/a}(X - T) + \theta\right] - \cos\left[\frac{2\pi}{\lambda/a}(X - Y)\right] \right) \quad [21]$$

where

$$C_0 = \frac{\pi E'}{2a} M^{0.5} L^{-0.5} (\gamma_1 - \beta_1) \bar{h} \quad [22]$$

and $C_0 = 0$ when an incompressible lubricant is assumed. M and L are the Moes parameters for either circular or elliptical contacts, as defined in the nomenclature. Perhaps the reader prefers to use instead of M the more meaningful parameter $\alpha p_H = (L/\pi)(\frac{3M}{2})^{1/3}$.

The Amplitude Reduction Curve

In order to complete the scheme for the calculation of pressures due to transverse sinusoidal waviness, Eq. [21] requires as input the amplitude of the convected film thickness (H_2). Here the amplitude reduction curve from Venner and Lubrecht (7) is used.

The amplitude of the lubricant film in the conjunction (A_d) is given as a function of the initial waviness amplitude (A_i) and operating conditions. Venner and Lubrecht give the following approximation for the amplitude ratio,

$$\frac{A_d}{A_i} = \frac{1}{1 + 0.15\bar{f}(r)\nabla + 0.015(\bar{f}(r)\nabla)^2} \quad [23]$$

where

$$\bar{f}(r) = \begin{cases} e^{1-\frac{1}{r}} & \text{if } r > 1 \\ 1 & \text{otherwise} \end{cases}$$

with $r = \lambda_x/\lambda_y$ and

$$\nabla = (\lambda/a)M^{0.5}L^{-0.5} \quad [24]$$

with $\lambda = \min(\lambda_x, \lambda_y)$.

Figure 1 depicts the amplitude reduction curve for both transverse and longitudinal waviness as given in Eq. [23]. The figure shows that for low frequencies (long wavelengths, $V \rightarrow \infty$) the roughness tends to be completely flattened (or simply over-rolled by the contact if its wavelength is about the contact size), where for high frequencies ($V \rightarrow 0$) the roughness passes through the contact unaltered. Still one must consider the definition of the parameter V , being not just a wavelength, but also reflecting the operating conditions of the contact.

The physical meaning of the V -parameter is the ratio between the wavelength of the waviness and the entrainment or inlet length. The entrainment length is defined by Hooke as "the distance from the dry contact end to the point where the maximum pressure gradient occurs."

The Amplitude of the Complementary Function

In pure rolling conditions the two waves travel with the same velocity and it is impossible to distinguish them. Besides, if the particular integral is small, it will seem as if the initial waviness of the gap is fully replaced by the convected wave. Therefore, the amplitude of the complementary function could be deduced directly from observations of the final shape in the center of the contact.

Greenwood (8) and Morales-Espejel, et al. (9) have used the amplitude reduction curve as the attenuation factor in the amplitude of the complementary function, $H_2 - Z_1 (A_d/A_i)$ and obtained reasonably good agreement with numerical solutions for dents and waviness. However, their analysis completely disregarded the particular integral. Hooke (10) shows the same scheme.

By a simple analysis of the amplitude of the particular integral as given by Eq. [14],

$$\frac{H_1}{Z_1} = \frac{C}{C + A} \quad [25]$$

one can clearly see that for very short wavelengths $A \rightarrow 0$ and therefore for compressible $C > 0$ fluids $H_1 \rightarrow Z_1$, the particular integral component remains *undeformed* and will certainly appear in the high pressure zone of the contact. The behaviour of Eq. [25] for short wavelengths seem to suggest that perhaps in reality the amplitude of the particular integral is reduced by second order effects for very short wavelengths. In any case, by taking the same assumptions as Greenwood (8) and Morales-Espejel, et al. (9), no difference in results for the pressures are observed for $\nabla \geq 0.3$ which covers most practical situations. For smaller values of V , the sum of the two components is greater than Z_1 (with a limit of $2Z_1$) and the pressure amplitude will be infinity, as also predicted by the Hooke's incompressible scheme Eq. [20].

Venner and Hooke predict that in pure rolling the final attenuation factor for the amplitude of the waviness (in the high pressure zone) has a limit of $A_d/A_i = 1$ for $V = 0$. Not even the most sophisticated numerical calculations would be able to prove this. The most recent calculations from Venner and Lubrecht (7) cannot get beyond say, $V = 0.25$.

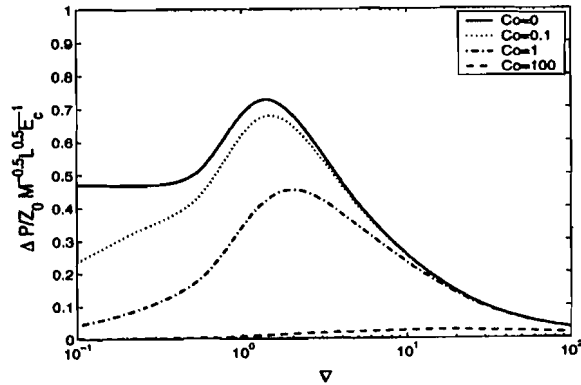


Fig. 2—Amplitude of the pressure ripples, according to Eqs. [21] and [23] for transverse waviness and different compressibility factors.

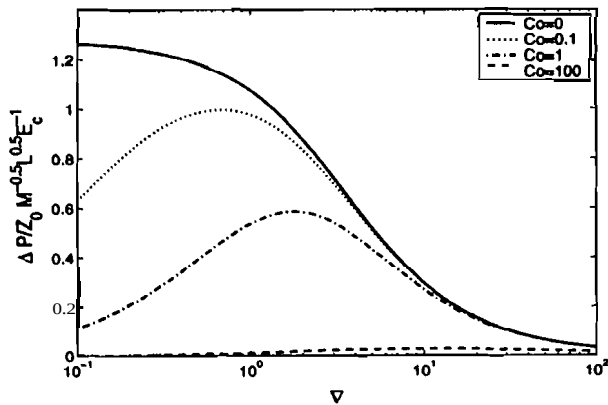


Fig. 3—Pressure perturbations as a function of V for longitudinal waviness in circular contacts, as predicted by Eq. [26], varying the compressibility.

If one accepts this limit, then the corresponding behaviour for the complementary function for the film thickness will not be the one described by the amplitude reduction curve from Venner, but the difference between this amplitude reduction curve and the particular integral, which of course, will be important for small values of V .

The amplitude reduction curve represents the behaviour of the amplitude for the final shape in the high pressure zone, so this curve must already have the two components. Of course, when sliding is present then it is not clear how one can measure this amplitude, since there will be at least two waves with different wavelengths, unless FFT techniques are used.

Figure 2 plots Eq. (21) with $H_2 = Z_1 (A_d/A_s)$ with the attenuation factor A_d/A_s as given by Eq. [23] for different values of the compressibility factor C_0 .

Pressures Due to Longitudinal Waviness Components

To the knowledge of the authors an analytical solution is not available for pressures due to longitudinal sinusoidal waviness. Nevertheless, for longitudinal waviness there is no inlet excitation and the relative velocities of the two surfaces does not change the results (the slide/roll ratio does not matter) then any steady-state

solution for longitudinal roughness can be used. The authors tried a slicing technique with steady-state line contacts along x , with central film thickness varying according to the waviness. However, the approach proved to be numerical and time consuming, thus it was abandoned. Instead, a simpler solution was used.

By observing the amplitude reduction curves from Venner and Lubrecht (7), shown in Fig. 1, it is easy to see that the transverse and longitudinal amplitude reduction curves follow a very similar trend. Besides, Eq. [23] shows that the control variable for the deformation of the roughness is $\bar{f}(r) V$, which becomes ∇_x (evaluated using ω_x) for transverse waviness and ∇_y (evaluated using ω_y) for longitudinal waviness. Full steady-state numerical solutions for longitudinal waviness were carried out, their pressure amplitudes have been plotted vs. ∇_y and it will be shown in the following section that they follow a single line. Indeed, if the amplitude reduction curves for transverse and longitudinal roughness follow similar patterns, one could also expect that the pressure ripple amplitude curves look similar. Therefore, any good curve-fit equation for longitudinal roughness must be similar to Eq. [21] for transverse roughness.

Knowing that the pressure perturbations particularly depend on the relative compressibility of the lubricant and the flexibility of the surface material, the following equation was found to correctly predict the amplitudes for pressure ripples due to longitudinal waviness,

$$M^{-0.5} L^{0.5} E_c^{-1} \frac{\Delta P_y}{Z_1} = \frac{\pi}{C_0 + \nabla_y} \left(\frac{\Delta H_y}{Z_1} - 1 \right) \quad [26]$$

The subscript y emphasizes the situation of longitudinal waviness. The pressure ripples can be expressed in terms of amplitudes, since in longitudinal waviness there is no influence of slip Lubrecht, et al. (18) and the pressures are in phase with the roughness.

Figure 3 shows amplitudes of the pressure perturbations as a function of V and C_0 .

Fourier Decomposition of a 3-D Surface

A three-dimensional irregular rough surface can be represented by a continuous function $z(x,y)$, where z is the vertical coordinate representing the surface height and x and y are orthogonal horizontal coordinates. This function $z(x,y)$ in signal processing stands for a two-dimensional signal depending on the independent variables x and y and can be regarded as a combination of series of sinusoidal waves, or Fourier series.

The discrete two-dimensional Fourier transform is given by a series of bi-sinusoidal waviness. This means that a surface roughness can be constructed by superimposing a series of elemental sinusoidal waves, defined by

$$z(x,y) = z_0 \cos(\omega_x x - \omega_y y) = z_1 \cos[\omega(x \cos \phi - y \sin \phi)] \quad [27]$$

The last function can be seen as an elemental component of 3-D roughness, its geometrical features controlled by the amplitude z_1 , the principal frequency $\omega = \sqrt{\omega_x^2 + \omega_y^2}$ and orientation angle $\phi = \tan^{-1}(\omega_y/\omega_x)$. The projection in x (transverse) and y (lon-

gitudinal) can be obtained by fixing $\phi = 0$ and $\phi = \pi/2$, respectively.

This decomposition in sinusoidal waves allows the use of any existing solution available for transverse and longitudinal sinusoidal waviness in EHL. Assuming linear behaviour one can calculate the pressure amplitude for every wavelength and orientation.

$$\frac{\Delta P}{Z_1} = \left(\frac{\Delta P}{Z_1}\right)_x \cos \phi + \left(\frac{\Delta P}{Z_1}\right)_y \sin \phi \quad [28]$$

The Complete Scheme

The 3-D pressures using a real measured surface can now be obtained by the following steps:

1. Apply FFT to the undeformed roughness to obtain its sinusoidal components for every frequency.
2. For every frequency ω_x apply Eq. [21] to calculate the corresponding pressure wave.
3. For every frequency ω_y apply Eq. [26] to calculate the amplitude of the corresponding pressure wave.
4. Apply Eq. [28].
5. Use IFFT to obtain the complete 3-D pressures.

SUBSURFACE STRESSES

Rolling contact fatigue, manifested as a flaking off of metallic particles of the surface, is known to depend on the interior stress field. Surfaces such as those of rolling contact bearings and gear teeth are subjected to loads which cause high subsurface stresses.

During rolling, the cyclic nature of these stresses often initiates cracks below the surface, which are propagated to the surface, eventually forming a pit or spall in the load carrying surface.

The load at which plastic yield begins is usually governed by von Mises' shear-stress criterion:

$$\frac{1}{6}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2] = k^2 = Y^2/3 \quad [29]$$

in which σ_1, σ_2 and σ_3 are the principal stresses in the state of complex stress and k and Y denote the values of the yield stress of the material in simple shear and simple tension (or compression) respectively.

In terms of Cartesian stresses the von Mises stresses in simple shear can be defined as

$$\tau_{\nu M} = \left\{ (1/6)[(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2 + 6(\tau_{xz}^2 + \tau_{yz}^2 + \tau_{xy}^2)] \right\}^{1/2} \quad [30]$$

The definition in simple tension (or compression) reads $\sigma_{\nu M} = \sqrt{3\tau_{\nu M}}$.

Roughness in an elastic contact will show two high value zones in the interior stress field. A maximum shear stress, analogous to the well studied Hertzian shear stress, is located well below the contact surface. The other zone is located very close to

the surface and the points of high stress are directly related to the surface roughness. Ai and Sawamiphakdi (19) have applied Fourier techniques before to the calculation of interior field stresses; they separated the pressures in the smooth surface (analytical solution available) and fluctuation component (solved by convolution). Here, convolution is not used. Instead, for the calculation the stresses due to the pressure fluctuation components, the harmonic waviness approach is used once again.

The assumption of an elastic half-space with a large thickness compared to the width of the loaded region, allows the 2-D stresses to be calculated using a plane strain approximation, comprising $\xi_y = \frac{\partial \nu_y}{\partial y} = 0$. So for a line contact the formulae for the subsurface stress due to a given pressure distribution $p = p_0 \cos(\alpha x)$, where α is the waviness-frequency, can be found from the stress function given in Johnson (20):

$$\phi_{x,z} = \frac{p_0}{\alpha^2} (1 + \alpha z) e^{-\alpha z} \cos(\alpha x) \quad [31]$$

Now the following expressions for the 2-D stress components can be obtained:

$$\begin{aligned} (\sigma_x)_p &= -p_0(1 - \alpha z) e^{-\alpha z} \cos(\alpha x) \\ (a,) &= -p_0(1 + \alpha z) e^{-\alpha z} \cos(\alpha x) \\ (\tau_{xz})_p &= -p_0(\alpha z) e^{-\alpha z} \sin(\alpha x) \end{aligned} \quad [32]$$

Morales-Espejel (4) and Hooke and Lee (21) successfully applied these equations to a fluctuating pressure distribution using Fourier decomposition.

Knowing that for a given surface traction $q = q_0 \cos(\alpha x)$, where q_0 might be μp_0 , the shear stress at $z = 0$ equals $\tau_{xz} = -q_0 \cos(\alpha x)$. The stress function for traction is then given by

$$\phi_{x,z} = \frac{q_0}{\alpha} z e^{-\alpha z} \sin(\alpha x) \quad [33]$$

Using this stress function leads to the stress components due to traction:

$$\begin{aligned} (a,) &= -q_0(2 - \alpha z) e^{-\alpha z} \sin(\alpha x) \\ (a,) &= -q_0(\alpha z) e^{-\alpha z} \sin(\alpha x) \\ (\tau_{xz})_q &= -q_0(1 - \alpha z) e^{-\alpha z} \cos(\alpha x) \end{aligned} \quad [34]$$

The Cartesian stresses, σ_x, σ_z and τ_{xz} can now be obtained by adding the components for normal pressure and traction.

If a tangential stress is applied at the contact surface in addition to the normal stress, the maximum von Mises stress will increase and move closer to the surface as the ratio of surface shear to normal stress increases. However, in pure rolling contacts μ is in general very small, thus surface tractions will be disregarded in the present analysis.

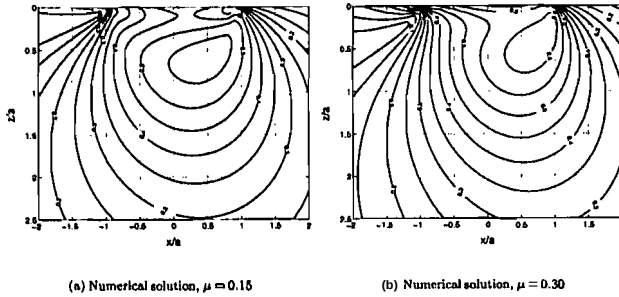
For a 3-D state of the interior stresses with a bi-sinusoidal pressure distribution on the surface, the equations can be obtained from Tripp, et al. (22).

TABLE 1—MAXIMA VON MISES STRESSES AND LOCATION FOR THE HERTZIAN LINE CONTACT PRESSURES

METHOD	μ	σ_{VM}/p_H	z/h
Analytical	0	0.557	0.73
2-D present scheme	0	0.557	0.7
3-D present scheme	0	0.560	0.7
Analytical	0.5	0.575	0.62
2-D present scheme	0.15	0.57	0.62
Analytical	0.30	0.666	0
2-D present scheme	0.30	0.670	0

TABLE 2—DIMENSIONLESS PRESSURE AMPLITUDES FOR ORIENTED AND BI-SINUSOIDAL WAVINESS

WAVINESS	$\phi = \pi/3$	$\phi = \pi/6$	BI-SINUSOIDAL
$\Delta p/p_0$ present	0.15	0.11	0.64

Fig. 4—von Mises stresses, σ_{VM}/p_H for a Hertzian pressure distribution with friction, using the 2-D present scheme.

The scheme has been validated using both periodic and non-periodic pressure distribution. For non-periodic pressures, an error caused by undesired periodic extension of the contact domain is implicitly introduced. This error is referred to as periodic extension error or periodicity error.

A very large window of 4 to 8 times the contact size is needed to reduce the magnitude of the periodic extension error, severely reducing the computational efficiency of the method. However, the present application will be used for the pressure fluctuation components only, these components have boundaries far away from the central zone of interest and can be assumed to be periodically repeated beyond these boundaries. If the macroscopic Hertzian stresses need to be added, analytical solutions can be used.

For validation purposes, Hertzian pressure distributions can be used together with a large window of zero-padding pressures. Figure 4 shows the results. To approximate the Hertzian pressures in a line contact problem using the 3-D scheme, a large window is used in the y direction. The numerical results obtained with the present scheme also are shown in Fig. 4. For $\mu = 0.15$ the subsurface stresses are shown in Fig. 4(a) and for $\mu = 0.3$ in Fig. 4(b), using a grid previously tested for convergence and a window of $8a \times 8a$. The analytical solution (plain strains) found in Johnson (20)

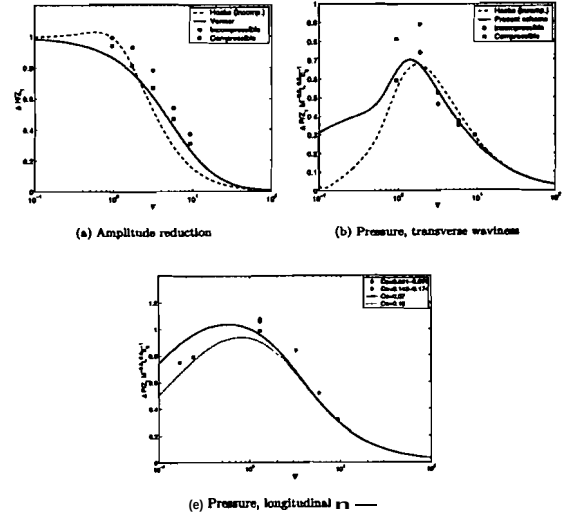


Fig. 5—Pressure and gap amplitudes from numerical solutions compared with the predictions of present theory for transverse and longitudinal waviness.

has been used as a comparison. The tabulated results for the maximum von Mises stress, τ_{VM}/p_H , and the matching depth, z/a , below the surface are given in Table 1 for the above examples and for two more examples with Hertzian pressures.

RESULTS

Surfaces With Sinusoidal Waviness

Full numerical solutions using multigrid techniques (described in e.g. Venner (3)) have been used to validate the model. Single sinusoidal surfaces with transverse, longitudinal, combined orientation, hi-sinusoidal and irregular pattern were used. The full set of data used are given Tables A1 and A2 in Appendix A.

The amplitude reduction curves for film thickness from Hooke (incompressible) and the approximation by Venner (compressible), Eq. [23], are compared with results from numerical simulations. The results can be seen in Fig. 5(a). One can see that Eq. [23] predicts the trend of the numerical results rather well. However, when incompressible results are added, the numerically predicted film ripple ratios for incompressible fluids are larger than the compressible ones for all values of V . If Hooke's incompressible curve is also added in the picture, one can see that the two curves cross more or less at $V = 2.5$, unlike the numerical results. In fact, the reason that incompressible fluids give higher amplitude ratios than compressible ones can be understood, since for the latter the deformation is the combined effect of the roughness deformation and the film reduction for compressibility. In incompressible fluids, it can be expected that the total deformed gap is smaller, since only the roughness is deformed. The crossing of the two curves is not yet understood.

The amplitude of pressure ripples for the transverse cases are given in Fig. 5(b) and for the longitudinal cases in Fig. 5(c). From these figures one can see that again the trend of the numerical results are correctly given by the present model, with a larger error in the central region of the curve for transverse roughness. In gen-

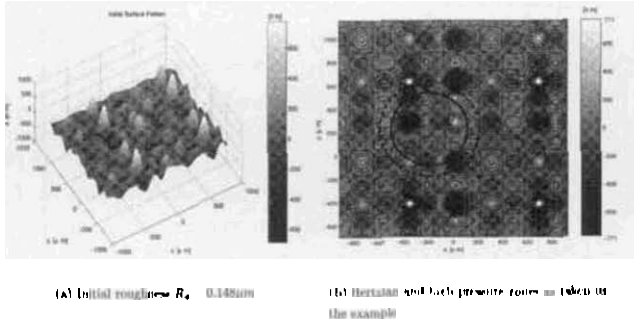


Fig. 6—Initial roughness with irregular pattern. (a) Initial roughness as generated by Eq. [35]. (b) Solid circle: Hertzian contact zone considered in the example. Dotted circle: high pressure zone (arbitrarily set as 70% of the contact zone radius).

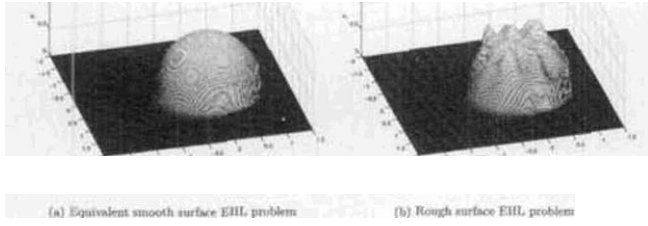


Fig. 7—Full numerical solutions of the EHL problem in dimensionless variables (129×129 points). (a) Equivalent smooth surface EHL solution. (b) Solutions for the present example with the roughness as described by Eq. [35].

eral compressible lubricants will produce larger pressure ripples (since the gap is more deformed).

Surfaces With Several Harmonics

The results from the solutions for the cases 19, 20 and 21 of Appendix A are given in Table 2 together with a comparison of result from numerical solutions. The first two cases are waviness with some orientation angle and the third one is a bi-sinusoidal waviness. The results are reasonably good specially for the last two cases.

For detailed verification of the model, a surface with a irregular pattern made of 5 harmonics with different phase angle and amplitude has been used, as described by the following formula,

$$z(x, y) = 0.1 \sum_{k=1}^5 \sum_{r=1}^5 e^{-\frac{k^2+r^2}{65}} \cos(6kx + \alpha_k) \cos(6ry + \beta_r) \quad [35]$$

with x, y in mm and z in microns $[\mu\text{m}]$. The phases are: $\alpha = [1 : 3 : 2 : 5 : 1]$ and $\beta = [0.5 : 1 : 4.3 : 1 : 2]$.

The operating conditions are the corresponding ones to example 24 from Table A2 (Appendix A). The R_q of the initial surface is $0.148 \mu\text{m}$. The generated 3-11 surface pattern is shown in Fig.

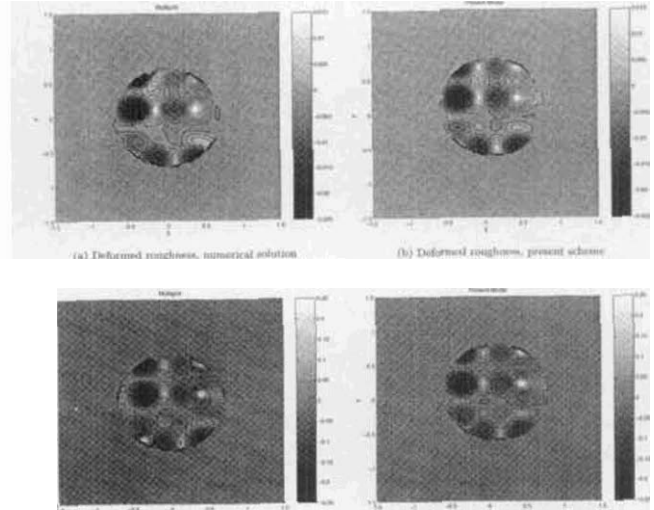


Fig. 8—Comparison in the high pressure zone of solutions for dimensionless deformed gap and pressure ripples between the present approach and the numerical (multigrid) method. Operating conditions from example 24.

6(a). Since the surface moves towards the contact in time (pure rolling along the x direction), the choice of a particular spot for the contact zone is equivalent to a selected snap shot in time. Figure 6(b) shows the selected spot for the 'Hertzian contact zone' (solid circle), the dotted circle shows what might be considered as the high pressure zone in the EHL contact for the equivalent smooth surface problem. here the radius is arbitrarily chosen as 70 % of that of the Hertzian circle.

Figure 7 depicts the full numerical solution for pressures in this example, obtained using the multigrid technique. Figure 7(a) shows the equivalent smooth surface solution and Fig. 7(b) the solution including roughness corresponding to the selected snap shot in time. This solution is now compared with the solution from the present analytical scheme.

In order to allow comparisons and since the present scheme deals only with the pressure ripples, the numerically-obtained smooth surface pressure distribution (Fig. 7(a)) has been subtracted from the rough surface pressure distribution (Fig. 7(b)) to obtain the corresponding 3-D pressure ripples, representative from the numerical solution. Figure 8(a) and 8(b) show the comparisons of the numerically-obtained deformed gap map and the present scheme. Figure 8(c) and 8(d) show the corresponding map of pressure ripples. For this example, the deformed surface resulted with $R_q = 0.086 \mu\text{m}$. One can observe that the agreement in form and values is very good, given the simplifications in the model. The maximum error found is located in the peaks and valleys close to the edges of the high pressure zone and it is about 25 % (higher values) for the deformed gap and 20 % (lower values) for the pressure ripples. Elsewhere the agreement is much better.

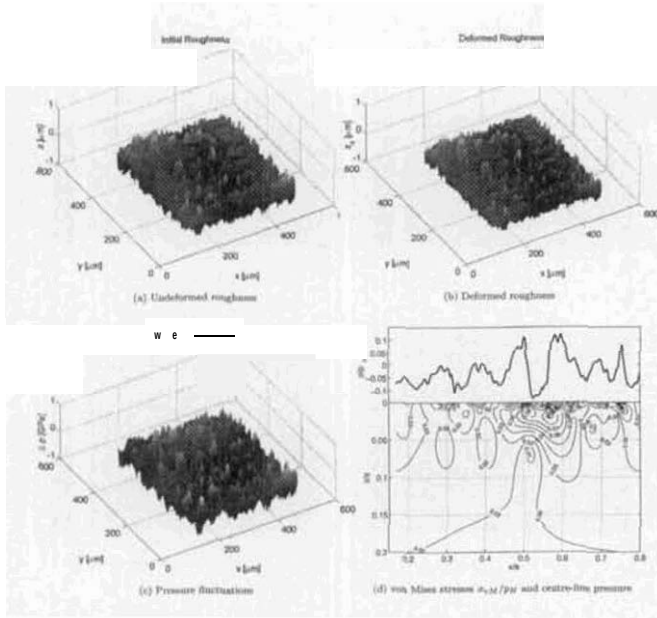


Fig. 9—Solutions for pressures, deformed gap and von Mises stresses using the present approach. Operating conditions and surface from example 22.

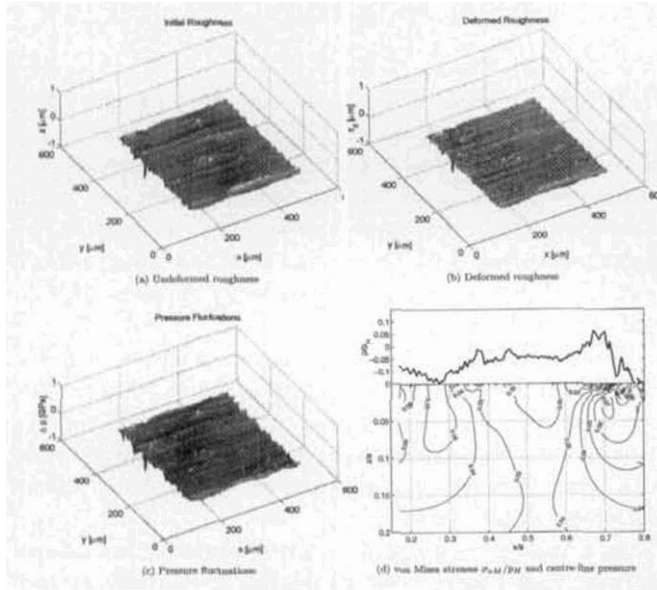


Fig. 10—Solutions for pressures, deformed gap and von Mises stresses using the present approach. Operating conditions and surface from example 23.

Surfaces With Real Roughness

Raceway surfaces from two rolling bearings of different size have been used as examples of real roughness. The operating conditions of the elliptical contact are given as examples 22 or 23 of Appendix A (same operating conditions). Samples of the two surfaces of $0.430 \text{ mm} \times 0.430 \text{ mm}$ have been measured using an optical profilometer. The first surface (example 22) with $R_q = 0.147 \mu\text{m}$ is shown in Fig. 9(a). Figure 9(b) shows the calculated

deformed shape ($R = 0.138 \mu\text{m}$) using the amplitude reduction curve, Eq. [23]. Fig. 9(c) shows the calculated pressure fluctuations to be expected in the center of the contact. Finally Fig. 9(d) shows the von Mises subsurface stresses corresponding to the total 3D pressure fluctuations. Comparing the vertical coordinate of Fig. 9 with that of Fig. 4, it is clear that Fig. 9 shows only the subsurface zone close to the surface, where the von Mises stress is dominated by the pressure ripples. In addition, on the top of the same figure the pressure along x in the center of the contact ($y = 0$) is shown. The maximum dimensionless von Mises stress corresponding to the pressure fluctuations for this example is $\tau_{\text{VM}}/p_H = 0.0961$. Figure 10 shows the corresponding plots for the second surface (example 23), with the same operating conditions as example 22. The initial surface has $R_q = 0.11711 \mu\text{m}$, the calculated deformed gap has $K = 0.064 \mu\text{m}$, which are lower values than the previous surface. The calculated maximum von Mises stress for this surface is $\tau_{\text{VM}}/p_H = 0.0729$. It is clear that surface 2 (example 23) produces lower pressure amplitudes than surface 1 (example 22). In this case it is mainly due to the lower R_q value, despite the nearly equal maximum height. However, it can be seen that the present scheme does not consider only one or a few roughness parameters but the whole geometry of the surface together with the rolling direction. In this way two real surfaces can be ranked.

DISCUSSION

In the present work, interesting aspects of the phenomena have been revealed and are further discussed below.

One important observation is the behaviour of the transverse component of the gap and pressure amplitudes for very small values of V . The pressure ripples will have a limiting value for $V \rightarrow 0$ that depends of the quantity $h_2 - z_1$.

For the case where the complementary function is assumed to replace completely the original shape of the surface (thus, the amplitude reduction curve represents the amplitude of the complementary function) then:

The incompressible analytical solution for transverse roughness 4, [20] in pure rolling with $\theta = 0$ should have a limiting value for the pressure amplitudes equal to zero, since the roughness will be completely undeformed and $h_2 = z_1$. In other words, it represents the limiting value of the amplitude reduction curve ($A_p/A = 1$). For the case of compressible lubricants, Eq. [18] shows that in the limit ($A \rightarrow 0$) when also $h_2 \rightarrow z_1$, the compressibility constant $C > 0$ makes the pressure amplitudes reach their zero limiting value faster.

This situation is not completely understood. However, a possible argument physically understandable is that, when the frequency increases, it becomes more difficult to deform the gap. In the limit the frequency is so high that no deformation is possible; therefore no pressure ripples can be generated. In the limit one comes back to a smooth surface situation.

Now, if the amplitude of the complementary function is taken to represent not one component but the final shape (i.e. the addition of the complementary function and the particular integral) then, in the limit, the particular integral is not deformed and the complementary function has to become zero in order to give the observed shape (amplitude reduction, with a limit equal to 1).

However, since $h_2 - z_1$ will now tend to Z , the predicted pressure ripples will have a limiting value different from zero, equal to $z_1/(C+A)$, as claimed in Morales-Espejel (4) and Greenwood and Morales-Espejel (5). So for incompressible fluids the limit will tend to infinity and for compressible ones it will tend to z_1/C . This behaviour now seems to be unrealistic.

One possible reason for the unrealistic behaviour of the second approach was given by Chang and Zhao (23). They removed the assumption of constant viscosity in the high pressures zone (contrary to the present scheme) and for the particular integral they obtained the same results except for very small values of V where they predict again a limiting value coming down towards zero.

As in Greenwood and Morales-Espejel (5), in order to retain the simplicity of the present scheme the assumption of constant viscosity in the high pressure zone is kept. Therefore, it is safer to limit the use of the model to $V \geq 0.3$ where the two variants still agree and numerical solutions exist to validate the amplitude reduction curve.

For longitudinal roughness, it was observed that the numerical solutions predict pressure amplitudes that also depend on V (calculated using the wavelength in they direction). Although in the present work no a rigorous explanation is offered for this, it is straightforward to see that according to the amplitude reduction curve, the deformation of any non-isotropic, transverse and longitudinal waviness is controlled by the parameter $\bar{f}(r)V$. With transverse roughness, this becomes ∇_x and with longitudinal waviness ∇_y . Therefore, the pressures should follow a similar dependency in either direction.

CONCLUSION

By means of the present understanding of sinusoidal waviness in EHL contacts and Fourier decomposition techniques, a full scheme has been developed for the calculation of the pressure fluctuations in rolling EHL contacts with real roughness. The scheme gives results in good agreement with numerical solutions for longitudinal, transverse, combined waviness direction and irregular patterns. The calculation of the interior stress field is also included. The method has useful engineering applications since it can be used to rank surfaces for their bearing performance. For illustration purposes, two different real surfaces in EHL contacts with different R_q values working under the same operating conditions have been ranked.

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REFERENCES

- (1) Lee, K. M. and Cheng, H. S. (1973). "Effects of Surface Asperity on Elastohydrodynamic Lubrication." NASA CR-2195. Washington, D.C.

- (2) Chang, L., Cusano, C. and Conry, T. F. (1989). "Effects of Lubricant Rheology and Kinematic Conditions on Micro-Elastohydrodynamic Lubrication," *ASME Jour. of Tribol.*, **111**, pp 344-351.
- (3) Venner, C. H. (1991). "Multilevel Solution of the EHL Line and Point Contact Problems," PhD Thesis, University of Twente. Enschede, The Netherlands.
- (4) Morales-Espejel, G. E. (1993), "Elastohydrodynamic Lubrication of Smooth and Rough Surfaces," PhD-Thesis. Engineering Department, University of Cambridge, U.K.
- (5) Greenwood, J. A. and Morales-Espejel, G. E. (1994). "The Behaviour of Transverse Roughness in EHL Contacts," in *Proc. Instn. Mech. Engrs., Part J*, **208**, pp 121-132.
- (6) Venner, C. H., Couhier, F., Lubrecht, A. A. and Greenwood, J. A. (1997), "Amplitude Reduction of Waviness in Transient EHL Line Contacts," in *Proc. of the 23th Leeds-Lyon Symp. on Tribol. 1996 Elsevier Tribology Series*, **32**, Dowson, D. et al., Eds., pp 103-112.
- (7) Venner, C. H. and Lubrecht, A. A. (1999). "Amplitude Reduction of Non-Isotropic Harmonic Patterns in Circular EHL Contacts, under Pure Rolling," in *Proc. of the 25th Leeds-Lyon Symp. on Tribol. 1998. Elsevier Tribology Series*, **34**, Dowson, D., et al., Eds., pp 151-162.
- (8) Greenwood, J. A. (1999), "Transverse Roughness in Elastohydrodynamic Lubrication," in *Proc. Instn. Mech. Engrs., Part J*, **213**, pp 383-396.
- (9) Morales-Espejel, G. E., Venner, C. H. and Greenwood, J. A. (2001), "Kinematics of Transverse Real Roughness in Elastohydrodynamically Lubricated Line Contacts Using Fourier Analysis," in *Proc. Instn. Mech. Engrs., Part J*, **214**, pp 523-534.
- (10) Hooke, C. J. (1998), "Surface Roughness Modification in Elastohydrodynamic Line Contacts Operating in the Elastic Piesoviscous Regime," in *Proc. Instn. Mech. Engrs.*, **212**, Pan J, pp 145-162.
- (11) Hooke, C. A. (1999). "The Behaviour of Low-Amplitude Roughness Under Line Contacts," in *Proc. Instn. Mech. Engrs.*, **213**, Pan J, pp 275-286.
- (12) Hooke, C. J. (2000). "The Behaviour of Low-Amplitude Surface Roughness Under Line Contacts: Non-Newtonian Fluids," in *Proc. Instn. Mech. Engrs.*, **214**, Pan J, pp 253-265.
- (13) Mnsen, M. A., Venner, C. H., Lugt, P. M. and Tripp, J. H. (2002), "Effects of Surface MicroGeometry on the Lift-Off Speed of an EHL Contact," *Tribol. Trans.*, **45**, **1**, pp 21-30.
- (14) Yuan-Zhong Hu, Barber, G. C., Zhu, D. and Ai, X. (2001), "A Study on an FFT-Based Approach for Fast Estimation of Pressure Distribution in Point EHL Contacts of Rough Surfaces," *Tribol. Trans.*, **44**, pp 59-69.
- (15) Lubrecht, A. A. and Ioannides, E. (1991), "A Fast Solution of the Dry Contact Problem and the Associated Sub-surface Stress Field. Using Multilevel Techniques," *ASME Jour. of Tribol.*, **113**, pp 128-132.
- (16) Hooke, C. J. and Venner, C. H. (2000). "Surface Roughness Attenuation in Line and Point Contacts," in *Proc. Instn. Mech. Engrs.*, **214**, Pan J, pp 439-444.
- (17) Venner, C. H., Kaneta, M. and Lubrecht, A. A. (2000), "Surface Roughness in Elastohydrodynamically Lubricated Contacts," in *Proc. of 26th Leeds-Lyon Symp. on Tribology. Leeds, 1999. Elsevier Tribology Series*, **38**, Dowson, D. et al. Eds., pp 25-36.
- (18) Lubrecht, A. A., ten Napel, W. E. and Bosma, R. (1988). "The Influence of Longitudinal and Transverse Roughness on the Elastohydrodynamic Lubrication of Circular Contacts," *ASME Jour. of Tribol.*, **110**, pp 421-426.
- (19) Ai, X. and Sawamiphakdi, K. (1999). "Solving Elastic Contact Between Rough Surfaces as an Unconstrained Strain Energy Minimization by Using CGM and FFT Techniques," *ASME Jour. of Tribol.*, **121**, pp 639-647.
- (20) Johnson, K. L. (1985). *Contact Mechanics*, Cambridge University Press.
- (21) Hooke, C. J. and Li, K. Y. (1999), "An Experimental Study of the Relationship Between Surface Roughness and Stress in EHL Contacts," in *Proc. 26th Leeds-Lyon Symp. on Tribol.*
- (22) Tripp, J., Kuilenburg, J. Morales-Espejel, G. E. and Lugt, P. M. (2002), "Subsurface Stress Formulas Useful in Rough Surface Stress Analysis," Submitted to *Tribol. Trans.*, STLE.
- (23) Chang, L. and Zhao, W. (1995), "Fundamental Differences Between Newtonian and Non-Newtonian Micro-EHL Results," *ASME Jour. of Tribol.*, **117**, pp 29-35.

APPENDIX A

Data Set for All the Examples Used

TABLE A1—DATA USED IN THE EXAMPLES

EXAMPLE	TYPE	$\frac{\alpha}{GPa^{-1}}$	$\frac{\eta_0}{Pa \cdot s}$	$\frac{E'}{GPa}$	$\frac{\mu H}{GPa}$	$\frac{\bar{h}}{\mu m}$	$\frac{u}{ms^{-1}}$	$\frac{R_x}{m}$	$\frac{a}{mm}$	$\frac{ z_{max} }{\mu m}$
1	transv.	22.0	0.04	220	1.4	0.55	3.0	0.014	0.24	0.110
	transv.	22.0	0.04	220	1.2	0.69	4.0	0.012	0.206	0.137
	transv.	22.0	0.04	220	0.95	0.59	3.0	0.012	0.16	0.118
4	transv.	22.0	0.04	220	2.0	0.54	3.0	0.014	0.40	0.108
5	transv.	14.8	0.04	220	2.0	0.42	2.5	0.018	0.51	0.084
6	transv.	11.5	0.03	220	2.0	0.31	2.5	0.020	0.57	0.062
7	long.	15.9	0.04	220	1.1	0.59	4.0	0.012	0.189	0.118
8	long.	15.9	0.04	220	1.13	0.58	4.0	0.012	0.194	0.058
9	long.	22.0	0.04	220	1.13	0.56	2.9	0.012	0.194	0.112
10	long.	22.0	0.04	220	2.0	0.62	4.0	0.012	0.343	0.124
11	long.	22.0	0.04	220	2.0	0.54	3.0	0.014	0.40	0.108
12	long.	14.8	0.04	220	2.0	0.42	2.5	0.018	0.51	0.084
13	long.	22.0	0.04	227	1.7	0.68	4.8	0.010	0.24	0.136
14	long.	22.0	0.04	227	1.2	0.53	3.0	0.010	0.17	0.106
15	long.	22.0	0.03	227	1.1	0.59	5.0	0.008	0.12	0.118
16	long.	11.5	0.03	220	2.0	0.31	2.5	0.020	0.57	0.062
17	long.	15.9	0.04	220	0.65	0.65	4.0	0.012	0.11	0.130
18	long.	22.0	0.04	220	0.95	0.59	3.0	0.012	0.16	0.118
19	$\phi = 60^\circ$	15.9	0.04	220	1.1	0.59	4.0	0.012	0.189	0.118
20	$\phi = 30^\circ$	18.2	0.04	220	1.4	0.50	3.0	0.012	0.240	0.099
21	$\phi = 0 - 90$	15.9	0.04	220	2.0	0.52	4.0	0.012	0.343	0.104
22	real	20.0	0.048	226	2.22	0.548	2.5	0.019	0.586	0.977
23	real	20.0	0.048	226	2.22	0.548	2.5	0.019	0.586	0.932
24	irregular	20.0	0.096	226	1.78	0.906	2.5	0.019	0.463	0.771

TABLE A2—DIMENSIONLESS PARAMETERS USED IN THE EXAMPLES

EXAMPLE	$z_{max}/\bar{h}$	λa	M	L	∇	C_0
1	0.2	0.5	14.9	180.93	1.74	0.114
2	0.3	0.4	16.1	91.8	0.956	0.143
3	0.2	0.2	14.9	56.5	1.74	0.174
4	0.2	0.5	14.4	592.2	3.21	0.072
5	0.2	0.6	8.68	819.8	5.83	0.066
6	0.2	0.7	6.11	1101	9.39	0.061
7	0.2	0.5	11.6	70.7	1.234	0.156
8	0.1	0.5	11.6	76.7	1.285	0.151
9	0.2	0.5	14.8	97.6	1.286	0.144
10	0.2	0.25	16.1	425.1	1.286	0.078
11	0.2	0.5	14.4	592.2	3.21	0.072
12	0.2	0.6	8.68	819.8	5.83	0.066
13	0.2	0.2	18.0	185.1	0.64	0.103
14	0.2	0.1	16.0	92.6	0.24	0.143
15	0.2	0.1	17.9	51.0	0.17	0.170
16	0.2	0.7	6.11	1101	9.39	0.061
17	0.2	0.5	11.6	14.6	0.56	0.256
18	0.2	0.2	14.9	56.5	0.39	0.174
19	0.2	0.5	11.6	70.7	$\frac{\nabla_x=0.617}{\nabla_y=1.069}$	0.156
20	0.2	0.5	12.4	180.9	$\frac{\nabla_x=1.657}{\nabla_y=0.957}$	0.114
21	0.2	1-0.5	11.6	425.1	$\frac{\nabla_x=6.052}{\nabla_y=3.026}$	0.077
22	1.782	-	959.55	12.37	-	0.0596
23	1.700	-	959.55	12.37	-	0.0596
24	0.851	-	290.95	14.77	-	0.0893